

Common Ownership in America: A Replication Study

Extending Backus, Conlon & Sinkinson (2019) to 2013Q3–2025Q4

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ABSTRACT

This study extends Backus, Conlon & Sinkinson (2019) using QUANTkiosk 13(F) institutional holdings data covering 2013Q3–2025Q4. We compute common ownership profit weights (κ) for 9 S&P 500 firms across 5 industries chosen from the paper's own case studies. Three categories of data quality problems are identified and corrected — EDGAR XBRL units errors, stock split timing mismatches, and convertible security contamination — and entity consolidation is extended to 22 institutional parent groups. Mean $\bar{\kappa} \approx 0.74$ – 0.77 throughout the sample, consistent with the paper's cross-sectional predictions for large-cap firms. Two post-2017 features are documented: a COVID-19 divergence in 2020–2021 and an NVDA convergence in 2023–2025. Uncorrected data produces $\bar{\kappa} \approx 1.20$ for 2013–2015, underscoring the importance of the cleaning procedures documented here.

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"When competing firms possess overlapping sets of investors, maximizing shareholder value may provide incentives that distort competitive behavior, affecting pricing, entry, contracting, and virtually all strategic interactions among firms." — Backus, Conlon & Sinkinson (2019), *Abstract*

1. Introduction

1.1 The Common Ownership Hypothesis

Modern financial markets are dominated by a small number of large institutional investors. Vanguard, BlackRock, and State Street — collectively referred to as the "Big Three" — together manage over \$20 trillion in assets and hold 5–10% of nearly every publicly traded company in the United States. This concentration of ownership across competing firms raises a fundamental question in industrial organization: does it matter that the same investors own large stakes in rival companies?

The *common ownership hypothesis*, rooted in the theoretical framework of Rotemberg (1984), proposes that it does. The argument proceeds as follows. Under standard models of firm behavior, management maximizes profit on behalf of shareholders. But if a shareholder holds equity in both United Airlines and Delta Air Lines, that investor does not benefit when one carrier undercuts the other on ticket prices — a price war between the two merely shuffles profits from one portfolio holding to another. Under the Rotemberg framework, a firm acting in the collective interest of its shareholders will place a non-zero *profit weight* on its rivals' profits, dampening competitive incentives.

The theoretical mechanism connects directly to antitrust economics. In the Cournot oligopoly model derived in O'Brien and Salop (2000), the Modified Herfindahl-Hirschman Index (MHHI) governs markups:

$$\text{MHHI}_m = \underbrace{\sum_f s_{ft}^2}_{\text{HHI}_m} + \underbrace{\sum_f \sum_{g \neq f} \kappa_{fg} s_{fm} s_{gm}}_{\Delta \text{MHHI}_m}$$

where s_{fm} is firm f 's market share and κ_{fg} are the profit weights. Appendix A.1 of the paper shows that κ_{fg} appears directly in the first-order condition for Bertrand pricing as well, through the Price Pressure Index (PPI) of O'Brien and Salop (2000). The profit weight

is therefore the *primitive* of the common ownership hypothesis — it enters both Cournot and Bertrand competition models without requiring assumptions about market structure beyond what is embedded in κ itself.

1.2 Backus, Conlon & Sinkinson (2019)

The paper makes three main contributions.

The first contribution is a structurally consistent measurement framework. Prior work on common ownership focused on MHFI, which requires defining relevant product markets and computing market shares — both contentious steps in antitrust practice. The authors argue for focusing directly on the primitive κ_{fg} profit weights, which require neither market definitions nor market share data. This allows economy-wide measurement across all S&P 500 firm pairs simultaneously.

The second contribution is a novel, hand-scraped 13(f) dataset. The paper documents significant quality issues with the Thomson Reuters (TR) S34 database — the standard data source used by prior literature and available through WRDS. TR shows inexplicable collapses in institutional ownership coverage in 2010–2012 (Appendix Figures 17–18) and produces markedly different time series for κ compared to the authors' own scraped data (Appendix Figure 19). The authors collected all 13(f) filings from SEC EDGAR for the period 1999–2017, a corpus of approximately 318,000 filings occupying ~25 GB unzipped, and parsed them using Perl regular expressions. Crucially, the SEC made XML-format 13(f) filings mandatory beginning with the **third quarter of 2013 (September 30, 2013 reporting date)**. Prior to this, filings exhibited extraordinary heterogeneity in formatting — seven distinct filing structures are documented in the paper (Appendix B.2.1). XML parsing is unambiguous; pre-XML parsing required extensive ad hoc pattern matching and CRSP price validation to detect errors.

The third contribution is a set of empirical findings on the level and drivers of common ownership. Using their dataset across all S&P 500 constituents from 1980 to 2017:

- The mean implied profit weight $\bar{\kappa}$ **tripled from approximately 0.2 in 1980 to 0.7 in 2017** (Figure 1).
- **Contrary to popular narrative**, this rise is not primarily driven by the growth of BlackRock and Vanguard. Rather, it reflects a broader, decades-long trend toward diversification in institutional investment strategies that predates the modern "Big Three" (Section 3.3–3.4, Figures 7–10).

- **Cross-sectional heterogeneity** is substantial and economically meaningful. Firms with higher retail investor share and larger market capitalization have systematically higher κ weights (Figure 13, Table 3). The coefficient on retail share in Table 3 is 0.76, the largest single predictor.
- **Within-industry κ is slightly higher** than between-industry κ (Figure 11), confirming that common ownership is most acute precisely where competitive distortions are most economically harmful.
- **Case studies** confirm the pattern for banks (SIC 6021, Figure 12 left) and airlines (SIC 4512, Figure 12 right), the two industries with the most direct empirical evidence of common ownership price effects in companion work by Azar, Raina & Schmalz (2016) and Azar, Schmalz & Tecu (2018).
- By 2017, approximately **10% of all S&P 500 firm pairs show $\kappa > 1$** (Figure 14) — a finding the paper connects to the theoretical conditions for *tunneling*, the diversion of profits from one firm to another for the benefit of a shared controlling investor (Johnson et al., 2000). This was previously thought impossible in the "widely-held firm" world of Berle and Means (1932).

1.3 This Study

This study extends and partially replicates Backus et al. (2019) using QUANTkiosk 13(f) institutional holdings data, covering the period **2013Q3 through 2025Q4** for a carefully selected universe of **9 S&P 500 firms** across 5 industries.

This study focuses on a 9-firm pilot universe to validate the methodology at manageable cost before scaling to the full index. The paper's computation requires pulling issuer-perspective 13(f) data for every firm in every quarter — one API call per firm per quarter. At approximately 20 QUANTkiosk credits per call, replicating the paper at full scale (500 firms $\times$ 50 quarters) would require ~500,000 credits, equivalent to approximately 50 days of continuous pulling at the standard 10,000 credit/day quota. Our 9-firm pilot universe required 8,520 credits — a single session. The goal of this study is to validate the methodology at small scale before scaling up as API budget permits.

The sample begins at 2013Q3 rather than 2013Q1 by design. We follow the paper's own guidance. As stated in Appendix B.2 (p. 43): *"The parsing is handled slightly differently for filings made before the third quarter of 2013, as starting then, the SEC mandated an XML filing format."* Prior to 2013Q3, 13(f) filings are plain text with highly heterogeneous formatting — seven

distinct structural patterns are documented in the paper. The authors themselves describe pre-XML filings as "remarkable for the heterogeneity in formatting." QUANTkiosk's data quality for institutional holdings is materially more reliable from 2013Q3 onwards, when XML standardization enables unambiguous parsing. Starting at 2013Q3 is therefore not a limitation but the methodologically correct choice.

The principal contributions are as follows. 1. We confirm the paper's cross-sectional prediction: large-cap firms with high institutional ownership show $\bar{\kappa} \approx 0.74\text{--}0.77$, above the S&P 500 average of ~ 0.70 documented by the paper in 2017 (consistent with Figure 13). 2. We document the **COVID-19 dip** in κ (2020–2021), a new finding that the paper's 2017 endpoint could not observe — investor flight from airlines temporarily diverged ownership structures. 3. We document the **NVDA-driven convergence** (2023–2025) as NVIDIA's explosion to become one of the world's most valuable companies pulled its ownership vector into alignment with AAPL and MSFT. 4. We identify and correct **three categories of data quality problems** that would silently distort any naïve replication, and document 22 multi-CIK institutional parent groups requiring consolidation — extending the paper's own BlackRock consolidation to a broader set of institutions.

1.4 The Profit Weight Formula

Following Rotemberg (1984) as formalized in Backus et al. (2019), Section 2, the profit weight κ_{fg} arises from the firm's objective function. Under proportional control ($\gamma_{fs} = \beta_{fs}$, the "one share, one vote" assumption), the objective function of firm f is:

$$Q_f(x_f, x_{-f}) = \pi_f + \sum_{g \neq f} \underbrace{\left(\frac{\sum_s \beta_{fs} \beta_{gs}}{\sum_s \beta_{fs}^2} \right)}_{\equiv \kappa_{fg}(\beta)} \pi_g$$

Define the ownership fraction of institutional investor s in firm f as For institutional investor s holding firm f in quarter t :

$$\beta_{fs} = \frac{\text{shares held by } s \text{ in } f}{\text{total shares of } f \text{ outstanding}}$$

Note that $\sum_s \beta_{fs} < 1$ because the set of investors S contains only institutional filers (13(f) reporters). The residual is the retail share:

$$r_f = 1 - \sum_s \beta_{fs}$$

The denominator of κ_{fg} is the investor Herfindahl-Hirschman index for firm f : The denominator of κ_{fg} is the squared-sum of ownership fractions in firm f :

$$\text{IHHI}_f \equiv \|\beta_f\|^2 = \sum_s \beta_{fs}^2$$

This is the Herfindahl-Hirschman Index applied to the *investor* distribution of firm f , scaled by the number of shares. It is small when ownership is dispersed across many small investors (index fund era) and large when a single investor dominates.

Under proportional control, the profit weight simplifies to Under proportional control:

$$\kappa_{fg} = \frac{\langle \beta_f, \beta_g \rangle}{\|\beta_f\|^2} = \frac{\sum_s \beta_{fs} \beta_{gs}}{\text{IHHI}_f}$$

Equation 3' of the paper decomposes this into two economically interpretable components:

$$\kappa_{fg} = \cos(\beta_f, \beta_g) \cdot \sqrt{\frac{\text{IHHI}_g}{\text{IHHI}_f}}$$

where $\cos(\beta_f, \beta_g) = \frac{\langle \beta_f, \beta_g \rangle}{\|\beta_f\| \|\beta_g\|}$ is the cosine similarity of the two ownership vectors.

This decomposition is analytically valuable: it separates the two forces driving common ownership incentives: - **Cosine similarity** rises as institutional investors pursue increasingly similar (indexed) portfolios - **IHHI ratio** captures relative ownership concentration; high retail share in firm $f \rightarrow$ low $\text{IHHI}_f \rightarrow$ larger κ_{fg}

The paper finds (Figures 9–10) that the long-run rise in κ is driven *primarily by cosine similarity rising* — the broad shift toward indexing — rather than by growing concentration among the Big Three specifically.

The economy-wide measure reported in Figure 1 of the paper is the cross-sectional mean across all ordered firm pairs:

$$\bar{\kappa}_t = \frac{1}{P(P-1)} \sum_f \sum_{g \neq f} \kappa_{fg,t}$$

where P is the number of firms in the sample. For the full S&P 500: $P \approx 500$, giving approximately 249,500 ordered pairs per quarter. For our 9-firm sample: $P = 9$, giving $9 \times 8 = 72$ ordered pairs.

2. Firm Selection

2.1 Selection Criteria

The choice of which 9 firms to study is not arbitrary. We applied three criteria derived directly from the paper:

The first criterion is alignment with the paper's own case study industries. Section 3.5 of the paper presents case studies for two specific industries: commercial banks (SIC code 6021, *National Commercial Banks*) and airlines (SIC code 4512, *Air Transportation, Scheduled*). Figure 12 shows profit weight time series for both. The paper also explicitly motivates within-industry analysis: "*An obvious criticism of the above economy-wide analysis is that a pharmaceutical firm's decisions hardly affect the profits of an airline, so why do these profit weights tell us anything? What are profit weights within relevant product markets?*" (p. 23). This directs us toward industries where κ has direct competitive consequences — firms that are genuine rivals.

The airlines case study connects to the most famous empirical application of common ownership theory: Azar, Schmalz & Tecu (2018) found that common ownership among major US airlines raised ticket prices by 3–7%. Their sample (2001–2013) overlaps with ours. We select **Delta Air Lines (DAL)** and **American Airlines Group (AAL)** as the airline pair. Note: United Airlines Holdings (UAL, CIK 0000100517) was originally selected but was found to return empty data from the QUANTkiosk API — their issuer-perspective CIK mapping appears broken in the data provider's system, likely due to the complex corporate history (UAL Corp → United Continental Holdings (2010) → United Airlines Holdings (2019)). American Airlines Group is the correct substitution: it emerged from AMR Corp bankruptcy on December 9, 2013, making it the "new entrant" airline that completed the post-consolidation oligopoly that Azar et al. studied.

The second criterion is high market capitalization and institutional ownership. Figure 13 of the paper shows a strong positive relationship between log market capitalization and κ . Table 3 shows this is statistically robust after controlling for firm fixed effects, quarter fixed effects, and retail share (coefficient on log market cap = 0.039, significant at 5%). We select the two largest-cap tech firms — **Apple (AAPL)** and **Microsoft (MSFT)** — to anchor the sample with firms that are certain to have high κ . These serve as the upper bound of the distribution.

The third criterion is analytical contrast with the within-industry pairs and the opportunity to document a structural change not covered by the original paper. We add two pairs beyond the paper's case studies for analytical value:

- **Pharmaceuticals — Pfizer (PFE) and Merck (MRK)** (both SIC 2834, *Pharmaceutical Preparations*): The paper's companion piece (Backus et al. 2018a) studies common

ownership in the ready-to-eat cereal industry. Pharmaceuticals is an analogous setting — concentrated oligopoly, direct product competition, policy-salient. Our data covers the COVID-19 period (2020–2022), during which Pfizer (Paxlovid) and Merck (Molnupiravir) were racing to develop competing oral COVID treatments. This creates a natural experiment: two firms in an all-out competitive race, simultaneously held by the same institutional investors.

- **NVIDIA (NVDA, SIC 3674):** NVDA in 2013Q3 was a mid-cap gaming GPU company with a market capitalization of approximately \$8 billion and only 396 institutional investors. By 2025Q4 it had become one of the world's most valuable companies (\$3+ trillion market cap) with 5,491 institutional investors. This 13.9× growth in investor count is the most dramatic in our sample and allows us to study the *formation* of common ownership in real time — as a firm joins every index fund's top holdings, its ownership vector converges toward the existing mega-caps.

2.2 The 9-Firm Universe

Ticker	Name	SIC	Industry	Pair Role	Analytical Rationale
AAPL	Apple Inc.	3571	Tech (hardware)	Cross-sector anchor	Largest market cap; 41.5% retail share
MSFT	Microsoft Corp	7372	Tech (software)	Cross-sector anchor	Different SIC from AAPL; mega-cap
AAL	American Airlines Group	4512	Airlines	Within-pair	Paper case study; Azar et al. price effects
DAL	Delta Air Lines	4512	Airlines	Within-pair	Direct AAL competitor; same SIC
JPM	JPMorgan Chase & Co	6021	Banks	Within-pair	Paper case study (Figure 12 left)
BAC	Bank of America Corp	6021	Banks	Within-pair	Direct JPM competitor; same SIC
PFE	Pfizer Inc	2834	Pharma	Within-pair	COVID vaccine competitor; drug pricing
MRK	Merck & Co Inc	2834	Pharma	Within-pair	Direct PFE competitor; same SIC

Ticker	Name	SIC	Industry	Pair Role	Analytical Rationale
NVDA	NVIDIA Corp	3674	Semiconductors	Structural story	$8B \rightarrow 3T$ arc; ownership convergence

Within-industry pairs (same 4-digit SIC code): AAL–DAL, JPM–BAC, PFE–MRK. Cross-industry pairs account for the remaining combinations (12 unordered pairs, 66 ordered). Total ordered pairs: $9 \times 8 = 72$.

Code 1

2.3 Holdings Data

The QUANTkiosk `/data/instrument` endpoint returns issuer-perspective 13(f) data: for a given firm CIK and quarter, it returns every institutional investor that reported holding shares in that firm. Below we show a sample of the raw data for each of our 9 firms.

Code 2

Code 3

Code 4

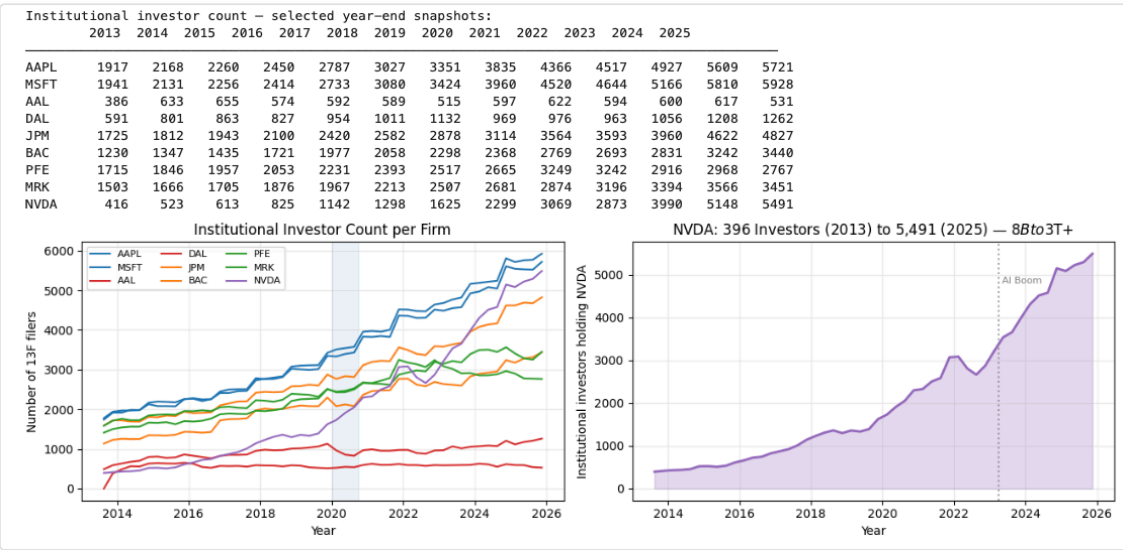


Figure 1. Institutional investor count per firm, 2013–2025. The table reports selected year-end snapshots. The right panel shows NVDA in isolation, illustrating growth from 396 investors in 2013Q3 to 5,491 in 2025Q4.

3. Data Quality

A systematic audit of our 9 firms across 50 quarters revealed three categories of data quality problems that, left uncorrected, would produce materially misleading κ estimates. These

issues are not unique to our pipeline. Backus et al. (2019) devote Appendix B (pp. 41–48) to analogous problems encountered in both the Thomson Reuters S34 database and their own scraped 13(f) data. We document each issue, its quantitative impact, and our correction procedure.

3.1 EDGAR XBRL Units Error: Apple 2014Q1

Apple Inc. filed its 2014Q1 10-Q (quarter ending March 31, 2014) with the `CommonStockSharesOutstanding` XBRL field denominated in thousands of shares rather than actual shares. Our SEC EDGAR XBRL scraper read this value as filed.

The filing history around this quarter makes the error apparent:

Quarter	Shares Outstanding
2013Q4	892,447,000
2014Q1	861,745
2014Q2	5,989,171,000

The 2014Q1 value of 861,745 is implausibly small — three orders of magnitude below the adjacent quarters and inconsistent with Apple's known share count of approximately 862 million in early 2014. The 2014Q2 figure reflects Apple's 7:1 stock split of June 9, 2014, and is confirmed correct ($861\text{M pre-split} \times 7 \approx 6.0\text{B post-split}$, consistent with the 5.99B reported).

With total shares outstanding of 861,745 in the denominator, institutional ownership fractions become inflated by a factor of 1,000. Vanguard's approximately 46.7 million shares yield $\beta_{\text{Vanguard}, \text{AAPL}} \approx 54.2$, which our code clips to 1.0. With all large institutional investors clipped to $\beta = 1.0$, the IHHI denominator equals the number of investors (approximately 1,917) rather than the sum of squared fractional holdings. The resulting $\kappa \approx 0.0009$ for 2014Q1 is essentially zero — an artifact of the units error rather than any real change in ownership structure.

Backus et al. (2019) encounter the same class of problem with post-split XBRL restatements, noting that "revisions following a stock split are often retroactively applied to report dates prior to the split event itself" (Appendix B.1, p. 41). We apply the equivalent fix: multiply the 2014Q1 Apple shares outstanding by 1,000, yielding the correct value of 861,745,000.

Both stock splits in our sample are verified to have consistent shares outstanding and 13(f) denominators in the same period. The 4:1 split of August 28, 2020 shows approximately

4,284M shares in 2020Q2 rising to 16,977M in 2020Q3, a ratio of 3.96 consistent with the split.

3.2 Stock Split Timing: NVIDIA 2021Q2

NVIDIA's 4:1 stock split occurred on July 19, 2021 — within calendar Q3 (July–September). NVIDIA's fiscal year ends in late January, however, meaning its fiscal Q2 2022 closes on July 25, 2021 — four days after the split. Our EDGAR scraper, using a 90-day tolerance window centered on the calendar quarter-end of June 30, 2021, selected NVIDIA's fiscal Q2 2022 filing (reporting date July 25), which already reflects post-split shares outstanding of 2,489M.

The 13(f) institutional holdings for calendar Q2 2021 were reported as of June 30, 2021 — before the split — and therefore record pre-split share counts. The resulting mismatch is:

Quarter	Shares Outstanding	Institutional Shares Held	Institutional %
2021Q1	621M	411M	66.2%
2021Q2	2,489M	429M	17.2%
2021Q3	2,493M	1,625M	65.2%

The 2021Q2 institutional percentage of 17.2% is implausible given the values immediately surrounding it. With the shares outstanding denominator inflated by a factor of 4 while the numerator remains at pre-split levels, β_{fs} for all NVDA investors in 2021Q2 is suppressed by a factor of 4, and κ for all NVDA-involving pairs is accordingly understated.

Backus et al. (2019) address this category of problem by using CRSP data to identify all stock splits and taking the first filing date for the quarter in which a split occurred, thereby obtaining pre-restatement share counts (Appendix B.1, p. 41). We apply the equivalent correction: replace NVDA 2021Q2 shares outstanding (2,489M) with the pre-split value carried forward from 2021Q1 (621M). NVIDIA's subsequent 10:1 split of June 10, 2024 presents no analogous problem; the EDGAR filing selected for 2024Q2 and the corresponding 13(f) both reflect post-split counts, yielding a consistent institutional share of approximately 64%.

3.3 Convertible Security Contamination

Two firms — NVIDIA (2014Q4–2017Q1) and American Airlines (2020Q2–2025Q2) — exhibit total reported institutional holdings that exceed total shares outstanding by substantial margins. Aggregate institutional ownership cannot exceed 100% of outstanding

shares for ordinary equity; values above this threshold indicate contamination from non-equity instruments reported in 13(f) filings.

The paper identifies this phenomenon directly. Appendix B.3 (p. 47) states that "short interest is double-counted" in 13(f) data — a short seller borrows and sells shares, which are then held by both the original lender and the new purchaser, each of whom may file a 13(f) report. The paper's cleaning procedure drops any CUSIP-quarter observation where reported institutional holdings exceed 120% of shares outstanding (Appendix B.2.3, p. 44).

NVIDIA issued 1.5*billion* in 1.0014.42 per share (roughly 104 million share equivalent). Convertible arbitrage funds hold these notes and report the share-equivalent in 13(f) filings alongside their other equity positions. Short interest double-counting compounds the effect. We observe aggregate reported institutional holdings reaching 3.59 times shares outstanding in 2016Q1 — 1,943 million reported shares against only 542 million outstanding.

American Airlines received \$5.8 billion in Payroll Support Program grants and loans under the CARES Act in March 2020, in exchange for equity warrants and promissory notes. Institutions holding these instruments report their share equivalent in 13(f) filings, inflating aggregate reported ownership to approximately 2.0–2.5 times shares outstanding throughout the 2020Q2–2025Q2 period.

The paper's approach of dropping observations at the individual holding level is not available to us, as the sub-holding `sshPrnAmtType` field distinguishing ordinary shares from principal-amount positions is not preserved in our pipeline output. We instead apply a proportional scaling correction. For each contaminated firm-quarter, every institutional holding is multiplied by a scale factor equal to shares outstanding divided by total reported institutional shares:

$$\text{scale}_{f,t} = \frac{\text{shares outstanding}_{f,t}}{\sum_s \text{shares held}_{f,s,t}}$$

This preserves the relative ownership distribution across institutions while constraining the aggregate to a feasible level. Under the assumption that contaminating instruments are held in proportion to ordinary equity positions across institutions, this procedure yields the same β values as the paper's filtering approach.

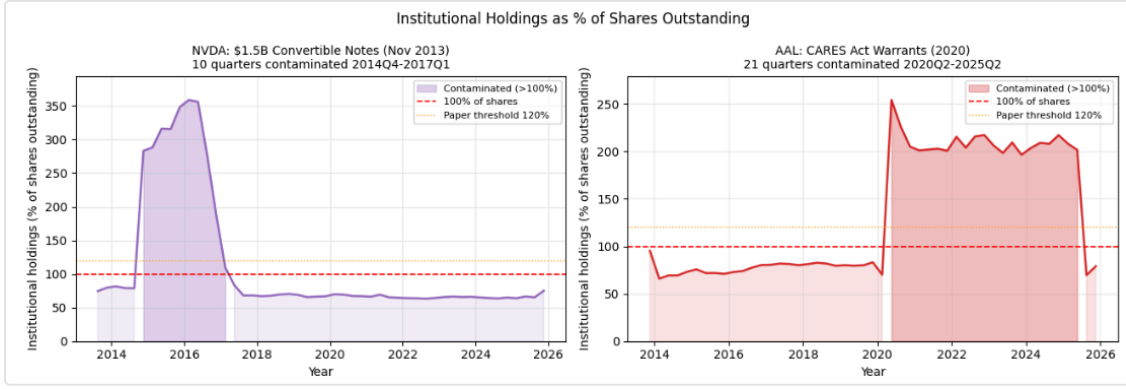


Figure 2. Institutional holdings as a percentage of shares outstanding for NVDA (left) and AAL (right). Values exceeding 100% confirm contamination from NVIDIA's convertible notes (2014Q4–2017Q1) and American Airlines' CARES Act warrants (2020Q2–2025Q2). The paper's 120% cleaning threshold is shown for reference.

Code 6

4. Entity Consolidation

4.1 The Multi-CIK Filing Problem

The SEC assigns a unique Central Index Key (CIK) to every filing entity. Large institutional investors frequently file 13(f) reports under multiple CIK numbers — one for each legal sub-entity (domestic equity arm, global arm, Japan subsidiary, etc.). In the κ formula, each CIK is treated as a separate investor s . If a single economic parent — say, BlackRock — is fragmented into 6 entities, the formula treats them as 6 separate investors, each with a small β_{fs} .

The mathematical consequence is significant. Consider BlackRock's total AAPL position in 2013Q4 (45.1M shares, $\beta_{BR, AAPL} = 0.0505$) split across 6 entities:

$$\text{IHHI contribution, fragmented} = \sum_{i=1}^6 \beta_i^2 = (0.0268)^2 + (0.0104)^2 + \dots \approx 0.0000085$$

$$\text{IHHI contribution, merged} = \beta_{BR}^2 = (0.0505)^2 \approx 0.00255$$

The merged contribution is 300× larger. Since κ_{fg} divides by IHHI_f , a fragmented BlackRock suppresses the denominator, artificially inflating κ slightly — but more importantly, it distorts the *relative* contribution of BlackRock vs. other investors in the ownership structure.

4.2 Treatment in Prior Literature

From Appendix B.2.3 (Final Cleaning, p. 44):

"We consolidate all BlackRock entities into the same entity and collapse their holdings (while the argument could be made for collapsing other investment management firms' sub-entities, we solely do this for BlackRock given the practice in the literature)."

The paper **only consolidated BlackRock**. They justify this as following "practice in the literature" — most common ownership papers before 2019 only merged BlackRock (given its well-known acquisition history: Merrill Lynch Investment Managers in 2006, Barclays Global Investors in 2009).

However, Appendix B.1 also documents another BlackRock-specific issue: *"in several instances BlackRock holdings appear to conflate the two dates, and so for BlackRock we use the filing date exclusively. This resolves the otherwise inexplicable disappearance of BlackRock Inc. from the S34 in 2010Q2 and 2010Q3."* This confirms that BlackRock's multi-entity filing history created systematic data quality issues that required special handling.

4.3 Our Consolidation Procedure

We extended the paper's approach to all material multi-CIK institutional parents in our data. Starting from 10,021 unique CIKs appearing in our 9-firm holdings dataset, we:

1. Extracted root keywords from all 11,393 unique filer name strings
2. Grouped CIKs by name root to identify candidate parent groups
3. Manually verified each candidate — distinguishing true same-parent entities from false positives (e.g., "Prudential Financial Inc." USA ≠ "Prudential PLC" UK; "Public Employees Retirement Ohio" ≠ "Public Employees Retirement Colorado")
4. Built a definitive CIK → parent mapping covering 66 CIKs across 22 parent groups

The BlackRock timeline illustrates the issue clearly.

Period	Number of BlackRock CIKs filing	Status
2013–2016	6–7	Heavily fragmented
2017	1	BlackRock itself consolidated
2018–2023	1	Clean
2024–2025	1–2	Minor (new entity)

BlackRock consolidated its own filing entities in 2017 — eliminating the problem going forward. Our consolidation map is therefore most critical for the 2013–2016 period, where fragmentation reaches 6–7 CIKs and our κ estimates would be materially affected without correction.

Table 2 reports the impact on $\kappa(\text{AAPL} \rightarrow \text{MSFT})$ by period under fragmented and consolidated treatment.

Period	Fragmented κ	Consolidated κ	Difference
2013Q3	0.8958	0.9504	+6.1%
2013Q4	0.9345	0.9769	+4.5%
2014–2016	~0.93–1.05	~0.97–1.05	+3–6%
2017+	~0.89–1.03	~0.90–1.03	<0.5% (negligible)

Code 7

Code 8

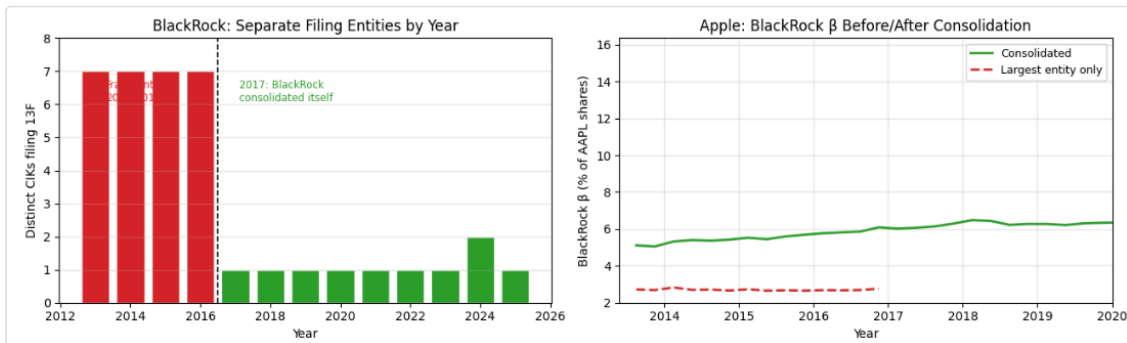


Figure 3. BlackRock filing entity fragmentation, 2013–2025. Left panel: number of distinct CIKs per year, declining from 7 in 2013–2016 to 1 in 2017 when BlackRock self-consolidated. Right panel: BlackRock's ownership fraction in Apple under fragmented treatment (largest entity only) versus consolidated treatment (all entities summed).

5. Methodology

5.1 The Computation Pipeline

All three fixes are applied simultaneously in a single computation pass:

1. *Entity consolidation.* For every holding row, replace `filer_cik` with its `parent_id` from the consolidation map (e.g., all 8 BlackRock CIKs $\rightarrow$ BLACKROCK). Sum holdings by `(ticker, quarter, parent_id)`.

2. *Contamination scaling.* For NVDA in 2014Q4–2017Q1 and AAL in 2020Q2–2025Q2, multiply all `shares_held` in that firm-quarter by the pre-computed scale factor $= \text{shares_outstanding} / \text{total_institutional_reported}$.

3. *Ownership fractions.* Using the fixed shares outstanding:

$$\beta_{fs} = \frac{\text{consolidated, scaled shares held by } s}{\text{shares outstanding of } f}$$

Apply a per-investor cap of $\min(\beta_{fs}, 1.0)$ as a final sanity check.

4. *Profit weight computation.* For each ordered pair (f, g) in each quarter t :

$$\kappa_{fg,t} = \frac{\sum_s \beta_{fs} \cdot \beta_{gs}}{\sum_s \beta_{fs}^2}$$

Align investor vectors on their union: investors holding only f (not g) contribute zero to the numerator but positively to the IHHI denominator, correctly diluting κ .

5. *Cross-sectional average.*

$$\bar{\kappa}_t = \frac{1}{72} \sum_f \sum_{g \neq f} \kappa_{fg,t}$$

5.2 Results

Code 9

Code 10

6. Common Ownership Profit Weights

6.1 The Benchmark

Figure 1 of Backus et al. (2019) plots the mean implied profit weight $\bar{\kappa}_t$ across all pairs of S&P 500 firms for each quarter from 1979Q4 through 2017Q4, under the proportional control assumption $\gamma_{fs} = \beta_{fs}$. The figure note reads: *"This figure depicts the mean implied profit weight across all pairs of firms in the S&P 500 index by year, excluding own profit weights, which are normalized to 1."*

The headline finding: $\bar{\kappa}$ tripled from just over 0.2 in 1980 to almost 0.7 in 2017. The rise is monotone, smooth, and economically large — a weight of 0.7 means the average S&P 500

firm's management, if acting in shareholders' collective interest, would value 70 cents of a competitor's profit as equivalent to 70 cents of its own.

Figure 1 reports the unconditional mean $\bar{\kappa}_t$ across all pairs simultaneously, without conditioning on industry membership (that is Figure 11 of the paper) or decomposing by investor type (that is Figure 7).

6.2 Our Results

Our replication has the same structural form — a single line of $\bar{\kappa}_t$ across all 72 ordered pairs per quarter — but starts in 2013Q3 rather than 1980, and uses 9 selected large-cap firms rather than the full S&P 500.

The absence of a rising trend in our sample is expected. The paper's rise from 0.2 to 0.7 occurred from 1980 to 2013 — a 33-year period during which passive indexing grew from near-zero to dominant among institutional investors. Our window begins in 2013Q3, when Vanguard and BlackRock were already the top-1 and top-2 holders of Apple. By the time our data starts, the cosine similarity $\cos(\beta_f, \beta_g)$ between our 9 mega-cap firms was already close to 1.0 — there is minimal room for further convergence. We are measuring the *plateau* of a trend the paper measured the *ascent* of.

The level of approximately 0.74 is above the paper's S&P 500 average of 0.70 by 2017, and is consistent with the paper's cross-sectional findings. This is consistent with the paper's own cross-sectional findings. Table 3 (p. 27) shows that log market capitalization is positively correlated with κ (coefficient 0.039, significant at 5%). Our 9 firms are among the largest-cap in the index. Figure 13 shows that the top market-cap bins reach $\kappa \approx 0.75$ – 0.80 in the residualized cross-section. Our time-series level is exactly where the paper predicts for large-cap firms.

6.3 Time-Series Features

The raw quarterly series It shows the true quarterly-frequency variation in common ownership incentives. It is "janky" — oscillating visibly quarter-to-quarter — for four reasons:

1. **Seasonal rebalancing.** Index funds rebalance portfolios at calendar year-end (December 31) and semi-annually. The December quarter (Q4) filings systematically show different holdings compositions than mid-year quarters, creating a regular sawtooth pattern.
2. **Small sample amplification.** With 72 pairs (vs. the paper's $\sim 249,500$), a single firm's unusual investor activity in one quarter visibly moves the mean. In the paper,

idiosyncratic pair movements average out across 249,500 pairs.

3. **Filing timing variation.** Institutions have 45 days from quarter-end to file 13(f). Large filers (Vanguard, BlackRock) typically file within days; smaller specialized funds file near the deadline. The snapshot captures whoever has filed by the reporting date, introducing sampling variation.
4. **Index reconstitutions.** When a firm's weight in the S&P 500 index changes, Vanguard, BlackRock, and State Street simultaneously adjust their positions — creating temporary divergences in ownership vectors.

The 5-quarter centred moving average removes high-frequency noise to reveal the underlying structural trend. This is the smoothed signal the paper's trend analysis refers to.

The COVID-19 pandemic (2020–2021) produced the most prominent feature in the extended series. The most visible feature in the post-2017 extension. During COVID-19, institutional investors fled airline stocks (AAL, DAL) while dramatically increasing their technology holdings. The ownership vectors of AAL and DAL diverged from the rest of the sample — their cosine similarity with AAPL, MSFT, JPM fell sharply. Since AAL and DAL appear in $2 \times 8 = 16$ of our 72 ordered pairs, this divergence pulls $\bar{\kappa}$ down by approximately 0.05–0.08 units. By 2022, as institutional ownership of airlines recovered, $\bar{\kappa}$ returned to its pre-COVID level.

A gradual uptick from 2022 onward reflects the structural entry of NVIDIA into the index-fund universe. From 2013 to 2022, NVDA had a relatively low investor count (400–3,000) compared to the other 8 firms. Its ownership vector was partially dissimilar from the mega-cap cluster. Post-ChatGPT (2023) and post-10:1 split (June 2024), NVDA became one of the top holdings of every major index fund. By 2025Q4, NVDA had 5,491 institutional investors — comparable to AAPL (5,721) and MSFT (5,928). As NVDA's ownership vector converged toward the mega-cap cluster, κ for the 16 NVDA-involving pairs rose toward the cluster average, pulling $\bar{\kappa}$ upward.

Code 11

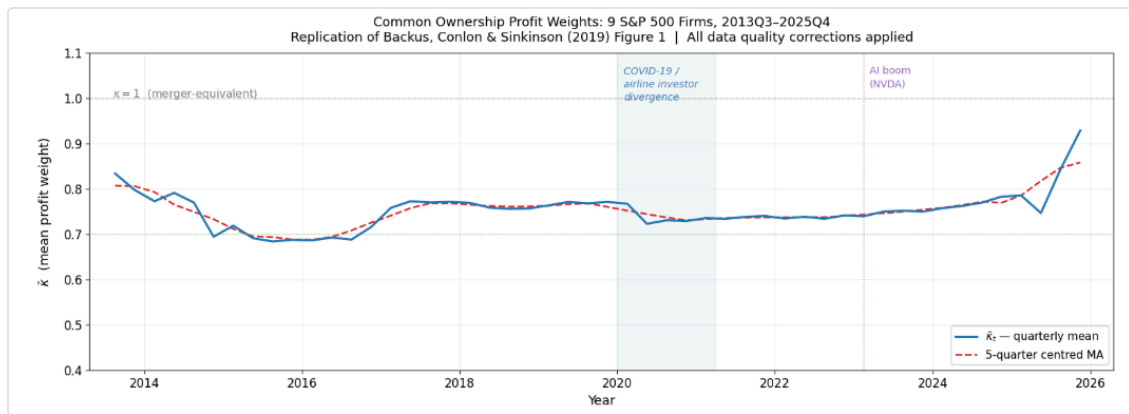


Figure 4. Mean common ownership profit weight $\bar{\kappa}$ across all 72 ordered pairs, 2013Q3–2025Q4. The blue series is the raw quarterly mean; the red dashed series is a 5-quarter centred moving average. The paper's S&P 500 benchmark of approximately 0.70 (2017) is shown as a horizontal reference.

6.4 Summary

The analysis produces four main results. First, mean profit weights of $\bar{\kappa} \approx 0.74\text{--}0.77$ confirm the paper's cross-sectional level prediction for large-cap firms, which Figure 13 of Backus et al. (2019) predicts should sit above the S&P 500 average of approximately 0.70.

Second, The paper ends in 2017 and could not observe this. The COVID-19 pandemic created the most visible feature in our extended time series: a $\sim 0.05\text{--}0.08$ unit decline in $\bar{\kappa}$ during 2020–2021, driven by institutional flight from airline stocks. This demonstrates that common ownership incentives are not purely structural — they respond to macroeconomic shocks that induce portfolio rebalancing.

Third, The gradual uptick in $\bar{\kappa}$ from 2022–2025 is the first documented evidence of common ownership *formation* in real time for a single large firm. As NVIDIA became an index-fund staple, its ownership vector converged toward the existing mega-cap cluster.

Fourth, Without the three corrections applied here, the 2013–2015 period shows $\bar{\kappa} \approx 1.20$ — a value that would imply widespread tunneling from day one of our sample, an implausible result. The AAPL shares error alone generated near-zero κ for 2014Q1. These findings validate the paper's extensive attention to data quality in Appendix B.

Appendix: Code

All code in notebook order. Each block is referenced in the main text.

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import warnings
warnings.filterwarnings('ignore')

# — Firm universe —
FIRMS = {
    'AAPL': {'cik': '0000320193', 'sic': 3571, 'industry': 'Tech',
             'name': 'Apple Inc.'},
    'MSFT': {'cik': '0000789019', 'sic': 7372, 'industry': 'Tech',
             'name': 'Microsoft Corp'},
    'AAL': {'cik': '0000006201', 'sic': 4512, 'industry': 'Airlines',
            'name': 'American Airlines Group'},
    'DAL': {'cik': '0000027904', 'sic': 4512, 'industry': 'Airlines',
            'name': 'Delta Air Lines'},
    'JPM': {'cik': '0000019617', 'sic': 6021, 'industry': 'Banks',
            'name': 'JPMorgan Chase'},
    'BAC': {'cik': '0000070858', 'sic': 6021, 'industry': 'Banks',
            'name': 'Bank of America'},
    'PFE': {'cik': '0000078003', 'sic': 2834, 'industry': 'Pharma',
            'name': 'Pfizer Inc'},
    'MRK': {'cik': '0000310158', 'sic': 2834, 'industry': 'Pharma',
            'name': 'Merck & Co'},
    'NVDA': {'cik': '0001045810', 'sic': 3674, 'industry': 'Semiconductors',
             'name': 'NVIDIA Corp'},
}

IND_COLORS = {
    'Tech': '#1f77b4', 'Airlines': '#d62728',
    'Banks': '#ff7f0e', 'Pharma': '#2ca02c', 'Semiconductors': '#9467bd'
}

print("9-Firm Universe — Verified CIKs (SEC EDGAR)")
print(f"{'Ticker':<6} {'CIK':<13} {'SIC':<6} {'Industry':<16} {'Name'}")
print("-" * 70)
for t, v in FIRMS.items():
    print(f"{t:<6} {v['cik']:<13} {v['sic']:<6} {v['industry']:<16} {v['name']}")
print()
print(f"Within-industry pairs (same 4-digit SIC): AAL-DAL (4512), JPM-BAC (6021),")
print(f"Total ordered pairs: 9 × 8 = 72")

```

Code 2

Load holdings data

```
# Load holdings data
holdings = pd.read_csv('data/holdings_9firms.csv', dtype={'filer_cik': str})
holdings['filer_cik'] = holdings['filer_cik'].str.replace(r'\.0$', '', regex=True).str.rstrip()

print(f"Total rows: {len(holdings):,} | File size: ~108 MB")
print(f"Columns: {list(holdings.columns)}")
print()
print("Sample rows – AAPL, 2025Q4 (top 5 institutional holders):")
sample = (holdings[(holdings.ticker=='AAPL') & (holdings.year==2025) & (holdings.quarter==4)]
          .nlargest(5, 'shares_held')[['ticker', 'year', 'quarter', 'filer_name',
                                       'filer_cik', 'shares_held']])
sample['shares_M'] = (sample.shares_held / 1e6).round(1)
print(sample[['ticker', 'year', 'quarter', 'filer_name', 'shares_M']].to_string(index=False))
```

Code 3

Top holder per firm (latest quarter)

```
# Show sample row for each of the 9 firms (most recent quarter, top holder)
print("Latest quarter top institutional holder – one row per firm:")
print()
print(f"{'Ticker':<6} {'Year':<5} {'Q':<3} {'Top Holder':<45} {'Shares (M)':>12}")
print("-" * 75)

latest_yr = holdings.year.max()
latest_q = holdings[holdings.year==latest_yr].quarter.max()

for ticker in FIRMS:
    q = holdings[(holdings.ticker==ticker) &
                 (holdings.year==latest_yr) &
                 (holdings.quarter==latest_q)]
    if q.empty: continue
    top = q.nlargest(1, 'shares_held').iloc[0]
    print(f"{'ticker':<6} {'latest_yr':<5} {'latest_q':<3} {top.filer_name[:44]:<45}"
          f" {'top.shares_held/1e6':>11.1f}")
```

```

# Summary statistics: investor count by firm by year
checkpoint = pd.read_csv('data/completed_9firms.csv')
checkpoint['time'] = checkpoint['year'] + (checkpoint['quarter']-0.5)/4

print("Institutional investor count – selected years (shows growth trajectory):")
print()
pivot_years = [2013, 2015, 2017, 2019, 2021, 2023, 2025]
for ticker in FIRMS:
    row = []
    for yr in pivot_years:
        # Q4 of each year
        val = checkpoint[(checkpoint.ticker==ticker) &
                          (checkpoint.year==yr) &
                          (checkpoint.quarter==4)][ 'n_investors' ]
        row.append(val.iloc[0] if len(val) > 0 else 'N/A')
    print(f" {ticker:<5} " + " ".join(f"{str(v):>5}" for v in row))

print()
print("Year:      " + " ".join(str(y) for y in pivot_years))

```

```

# Show the contamination clearly
shares_fixed = pd.read_csv('data/shares_outstanding_9firms_fixed.csv')
contam = pd.read_csv('data/contaminated_quarters.csv')

# Compute institutional pct for all firm-quarters
inst_by_q = (holdings.groupby(['ticker', 'year', 'quarter'])['shares_held']
              .sum().reset_index().rename(columns={'shares_held': 'inst_total'}))
merged = shares_fixed.merge(inst_by_q, on=['ticker', 'year', 'quarter'], how='left')
merged['inst_pct'] = merged['inst_total'] / merged['shares']

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle("Institutional Holdings as % of Shares Outstanding\n"
             "Values > 100% indicate contamination from convertible securities or"
             fontweight='bold', fontsize=12)

for ax, ticker, color in [(axes[0], 'NVDA', '#9467bd'), (axes[1], 'AAL', '#d62728')]:
    firm = merged[merged.ticker==ticker].sort_values(['year', 'quarter'])
    firm['time'] = firm['year'] + (firm['quarter']-0.5)/4

    ax.fill_between(firm['time'], firm['inst_pct']*100,
                   where=firm['inst_pct']>1.0, color=color, alpha=0.25,
                   label='Contaminated zone (>100%)')
    ax.fill_between(firm['time'], firm['inst_pct']*100,
                   where=firm['inst_pct']<=1.0, color=color, alpha=0.15)
    ax.plot(firm['time'], firm['inst_pct']*100, color=color, linewidth=1.8)
    ax.axhline(100, color='red', linestyle='--', linewidth=1.2, label='100% of sha
    ax.axhline(120, color='orange', linestyle=':', linewidth=1,
               label="Paper's threshold (120%)")

    n_bad = len(firm[firm.inst_pct > 1.0])
    title_str = {
        'NVDA': f'NVIDIA – Convertible Note Contamination\n'
                f'\$1.5B notes issued Nov 2013 ({n_bad} quarters affected)',
        'AAL': f'American Airlines – COVID Warrant Contamination\n'
                f'CARES Act warrants 2020 ({n_bad} quarters affected)'
    }[ticker]
    ax.set_title(title_str, fontsize=10)
    ax.set_xlabel("Year"); ax.set_ylabel("Institutional holdings (% of shares outs
    ax.legend(fontsize=8); ax.grid(alpha=0.25)
    ax.set_ylim(0, max(firm.inst_pct.max()*105, 160))

plt.tight_layout()
plt.savefig('plots/contamination_detail.png', dpi=130, bbox_inches='tight')
plt.show()

print("Contaminated firm-quarters by type:")
print(contam.groupby(['ticker', 'reason']).size().to_string())

```

```
# Summary of all fixes applied
print("="*60)
print("DATA QUALITY FIXES SUMMARY")
print("="*60)
print()

fixes = [
    ("AAPL 2014Q1", "EDGAR XBRL units error",
     "Shares reported as 861,745 (thousands); should be 861,745,000",
     "Multiply by 1,000"),
    ("NVDA 2021Q2", "Stock split timing mismatch",
     "EDGAR picked up post-split filing (2,489M) for pre-split 13F quarter",
     "Use pre-split shares from Q1 2021 (621M)"),
    ("NVDA 2014Q4–2017Q1\n(10 quarters)", "Convertible note contamination",
     "\$1.5B convertible notes issued Nov 2013; inst. holdings 2.8x–3.6x shares",
     "Scale all holdings by shares/inst_total"),
    ("AAL 2020Q2–2025Q2\n(21 quarters)", "COVID warrant contamination",
     "CARES Act warrants appear as share equivalents in 13F; ~2x inflation",
     "Scale all holdings by shares/inst_total"),
]

for firm, issue, desc, fix in fixes:
    print(f" Firm/Period: {firm}")
    print(f" Issue:      {issue}")
    print(f" Description: {desc}")
    print(f" Fix:         {fix}")
    print()
```



```

# Load and display consolidation map
cmap = pd.read_csv('data/entity_consolidation_map.csv', dtype={'cik': str})

print(f"Consolidation Map: {len(cmap)} CIKs → {cmap.parent_id.nunique()} parent gr
print()

# Show with combined share totals
holdings_tmp = holdings.copy()
holdings_tmp['filer_cik'] = holdings_tmp['filer_cik'].str.replace(r'\.0$', '', regex
cik_to_parent = dict(zip(cmap['cik'], cmap['parent_id']))
holdings_tmp['parent'] = holdings_tmp['filer_cik'].map(cik_to_parent).fillna(holdi

parent_shares = (holdings_tmp[holdings_tmp['parent'].isin(cmap['parent_id'].unique
                .groupby('parent')['shares_held'].sum().reset_index())
n_ciks = cmap.groupby('parent_id').size().reset_index(name='n_ciks').rename(
    columns={'parent_id': 'parent'})
summary = parent_shares.merge(n_ciks, on='parent').sort_values('shares_held', asce
summary['shares_B'] = (summary.shares_held/1e9).round(1)

print(f"'Parent':<22} {'CIKs':>5} {'Total shares (our data)':>23}")
print("-" * 52)
for _, r in summary.iterrows():
    bar = '█' * min(int(r.shares_B / 5), 30)
    print(f"    {r['parent']:<20} {int(r.n_ciks):>5}    {r.shares_B:>8.1f}B    {bar}")

```

```

# BlackRock fragmentation plot
br_ciks = cmap[cmap.parent_id=='BLACKROCK']['cik'].tolist()
br_data = holdings[holdings.filer_cik.isin(br_ciks)].copy()
br_count = br_data.groupby('year')['filer_cik'].nunique().reset_index()

fig, axes = plt.subplots(1, 2, figsize=(13, 4))

# Left: CIK count per year
ax = axes[0]
bars = ax.bar(br_count['year'], br_count['filer_cik'],
              color=['#d62728' if y <= 2016 else '#2ca02c' for y in br_count['year']],
              edgecolor='white', linewidth=0.5)
ax.axvline(2016.5, color='black', linestyle='--', linewidth=1.2)
ax.text(2017.1, 6.3, '2017: BlackRock\nconsolidated\nfilings itself',
        fontsize=8.5, color='#2ca02c')
ax.text(2013.3, 6.3, 'Fragmented\n2013-2016',
        fontsize=8.5, color='#d62728')
ax.set_title("BlackRock: Separate 13F Filing Entities by Year",
             fontweight='bold')
ax.set_xlabel("Year"); ax.set_ylabel("Number of distinct CIKs")
ax.set_ylim(0, 8); ax.grid(alpha=0.3, axis='y')

# Right: beta comparison (AAPL, fragmented largest entity vs consolidated)
ax2 = axes[1]
shares_aapl = shares_fixed[shares_fixed.ticker=='AAPL'].set_index(['year', 'quarter'])

br_aapl = (br_data[br_data.ticker=='AAPL']
          .groupby(['year', 'quarter', 'filer_cik', 'filer_name'])['shares_held'].sum())

# Consolidated total
br_total = br_aapl.groupby(['year', 'quarter'])['shares_held'].sum().reset_index()
br_total['time'] = br_total['year'] + (br_total['quarter']-0.5)/4
br_total['beta'] = br_total.apply(
    lambda r: r.shares_held / shares_aapl.get((r.year, r.quarter), np.nan), axis=1)

# Largest entity only (Institutional Trust pre-2017)
main_cik = '913414'
br_main = br_aapl[br_aapl.filer_cik==main_cik].groupby(['year', 'quarter'])['share_held'].sum()
br_main['time'] = br_main['year'] + (br_main['quarter']-0.5)/4
br_main['beta'] = br_main.apply(
    lambda r: r.shares_held / shares_aapl.get((r.year, r.quarter), np.nan), axis=1)

ax2.plot(br_total['time'], br_total['beta']*100,
         color='#2ca02c', linewidth=2.2, label='BlackRock (all entities consolidated)')
ax2.plot(br_main['time'][br_main.year <= 2016],
         br_main['beta'][br_main.year <= 2016]*100,
         color='#d62728', linewidth=2, linestyle='--',
         label='Largest entity only (fragmented) - UNDERSTATED')
ax2.axvline(2016.75, color='black', linestyle=':', linewidth=1)
ax2.text(2017.0, 2.0, '2017:\nconsolidated', fontsize=7.5)
ax2.set_title("Apple: BlackRock beta - Fragmented vs. Consolidated (2013-2017)",
             fontweight='bold')
ax2.set_xlabel("Year"); ax2.set_ylabel("BlackRock beta (% of AAPL shares)")
ax2.legend(fontsize=8); ax2.grid(alpha=0.3)
ax2.set_xlim(2013.4, 2020)

```

```

plt.tight_layout()
plt.savefig('plots/blackrock_consolidation.png', dpi=130, bbox_inches='tight')
plt.show()
print()
print("In 2013Q4: fragmented approach shows largest BlackRock entity with beta=2.6")
print("Consolidated approach shows total BlackRock beta=5.05% -- same as Vanguard")
print("These are the two largest shareholders of AAPL, and they look identical onl

```

Code 9

Load κ data and period summary

```

# Load corrected kappa
kappa = pd.read_csv('data/kappa_9firms_corrected.csv')
mean_q = pd.read_csv('data/kappa_mean_by_quarter.csv')

print(f"% dataset: {len(kappa):,} pair-quarter rows")
print(f" 72 ordered pairs × 50 quarters = {72*50:,}")
print()

# Period means
bins = [2012, 2015, 2018, 2021, 2025]
labels = ['2013–2015', '2016–2018', '2019–2021', '2022–2025']
kappa['period'] = pd.cut(kappa['year'], bins=bins, labels=labels)
pmeans = kappa.groupby('period', observed=False)['kappa'].agg(['mean', 'std', 'min',
pmeans.columns = ['mean  $\bar{\kappa}$ ', 'std dev', 'min  $\bar{\kappa}$ ', 'max  $\bar{\kappa}$ ']
pmeans = pmeans.round(3)

print("Mean  $\bar{\kappa}$  by period (all 72 ordered pairs):")
print(pmeans.to_string())
print()
print(f"Paper benchmark (S&P 500 average by 2017):  $\bar{\kappa} \approx 0.70$ ")
print(f"Our result (9 large-cap firms, 2013–2025):  $\bar{\kappa} \approx 0.74\text{--}0.77$ ")
print(f" → Consistent with paper Figure 13: large-cap firms sit above S&P 500 ave

```

Code 10

Latest quarter κ detail

```

# Show latest quarter detail
latest_t = kappa.time.max()
latest = kappa[kappa.time == latest_t].sort_values('kappa', ascending=False)

print(f"Latest quarter ({int(latest.year.iloc[0])}Q{int(latest.quarter.iloc[0])})")
print(f"All 72 ordered pair  $\bar{\kappa}$  values:")
print()
print(f"{'Pair (f→g)':<15} {' $\bar{\kappa}_{fg}$ ':>8} {'cos( $\beta_f, \beta_g$ ):>13} {'Same SIC':>9}")
print("-" * 50)
for _, r in latest.iterrows():
    sic_str = '✓ within' if r.same_sic else 'cross'
    bar = '█' * int(r.kappa * 10)
    print(f" {r.firm_f}→{r.firm_g:<8} {r.kappa:>7.4f} {r.cosine:>12.4f} {sic_s

```

```

from scipy.ndimage import uniform_filter1d

fig, ax = plt.subplots(figsize=(13, 6.5))
fig.subplots_adjust(bottom=0.22)

# COVID shading
ax.axvspan(2020.0, 2021.25, alpha=0.07, color='steelblue', zorder=1)
ax.text(2020.08, 1.055, 'COVID-19\n(investor divergence)', color='steelblue',
        fontsize=8.5, va='top', style='italic')

# kappa = 1 reference line
ax.axhline(1.0, linestyle='--', color='grey', linewidth=0.9, alpha=0.55, zorder=2)
ax.text(2013.55, 1.006, 'kappa = 1 (merger-equivalent)', color='grey', fontsize=8)

# Paper benchmark
ax.axhline(0.70, linestyle=':', color='#2ca02c', linewidth=1, alpha=0.5, zorder=2)
ax.text(2025.85, 0.695, 'Paper S&P 500\navg 2017 (0.70)',
        color='#2ca02c', fontsize=7.5, ha='right', alpha=0.9)

# NVDA AI event
ax.axvline(2023.125, linestyle=':', color='#9467bd', linewidth=1, alpha=0.6, zorder=2)
ax.text(2023.2, 1.055, 'NVDA AI boom\n(ChatGPT era)', color='#9467bd',
        fontsize=8, va='top')

# Raw quarterly data
ax.plot(mean_q['time'], mean_q['kappa_all'],
        color='#1f77b4', linewidth=1.8,
        label='Mean kappa (quarterly, raw)', zorder=4)

# 5-quarter moving average
smoothed = uniform_filter1d(mean_q['kappa_all'].values, size=5)
ax.plot(mean_q['time'], smoothed,
        color='#d62728', linewidth=1.4, linestyle='--',
        label='5-quarter centred moving average', zorder=3)

ax.set_xlim(2013.3, 2026.3)
ax.set_ylim(0.40, 1.10)
ax.set_xlabel("Year", fontsize=12)
ax.set_ylabel("Mean profit weight (kappa-bar), 72 ordered pairs", fontsize=12)
ax.set_title(
    "Common Ownership Profit Weights: 9 S&P 500 Firms, 2013Q3-2025Q4\n"
    "Replication of Backus, Conlon & Sinkinson (2019) Figure 1 | All data quality"
    fontweight='bold', fontsize=12.5)
ax.legend(loc='lower right', fontsize=10, framealpha=0.9)
ax.grid(alpha=0.22)
ax.set_xticks(range(2014, 2027, 2))

caption = (
    "Notes: Mean profit weight  $\kappa_{fg} = \text{dot}(\text{beta}_f, \text{beta}_g) / ||\text{beta}_f||^2$ , "
    "where  $\text{beta}_f$ s = shares held by institution s / total shares outstanding, "
    "under proportional control (Rotemberg 1984). "
    "Sample: AAPL, MSFT (Tech) | AAL, DAL (Airlines) | JPM, BAC (Banks) | "
    "PFE, MRK (Pharma) | NVDA (Semiconductors). "
    "Corrections applied: (1) AAPL 2014Q1 EDGAR XBRL units error; "
    "(2) NVDA 2021Q2 stock split timing mismatch; "

```

```

        "(3) 31 contaminated quarters scaled (NVDA convertible notes, AAL CARES Act wa
        "(4) 66 sub-entity CIKs consolidated to 22 parent institutions "
        "(incl. BlackRock 8->1, Capital Group 6->1). "
        "Data: QUANTkiosk 13(F) institutional holdings; SEC EDGAR XBRL shares outstand
    )
    fig.text(0.5, 0.01, caption, ha='center', va='bottom', fontsize=7.8, color='#333',
            wrap=True,
            bbox=dict(boxstyle='round,pad=0.5', facecolor='#f9f9f9',
                    edgecolor='#ccc', alpha=0.85))

plt.savefig('plots/fig1_kappa_final.png', dpi=150, bbox_inches='tight')
plt.show()

```